

Hudson C. Smith

2315 Wallace St.
Philadelphia, PA 19130

Email: hudsoncsmith@gmail.com
Phone: 215-300-2206

Education

Master of Science, Mechanical Engineering - Virginia Tech, Blacksburg, VA, 2023

Master of Science, Mechanical and Process Engineering - TU Darmstadt, Darmstadt, Germany, 2023

Bachelor of Arts, Physics / German Studies - Hamilton College, Clinton, NY, 2021

Professional Experience

Mechanical Engineering Designer II, GreenbergFarrow Architecture New York, NY Dec 2023 – Present

- Assisting with updates to prototypical Mechanical, Electrical, and Plumbing (MEP) systems for Murphy Oil convenience stores across the United States and taking ownership of the MEP design for individual stores
- Performing advanced drafting, equipment selection and system analysis in Autodesk Revit
- Coordinating with architects, structural and civil engineers, and clients to ensure that design needs are met
- Visiting active construction sites to ensure that projects are built according to design specifications

Working Thesis Student, ABB Robotics – Corporate Research Ladenburg, DE May 2023 – Nov. 2023

- Authored a seminar paper on a machine learning application for safe robot stopping behavior in the German mechatronics conference VDI Mechatroniktagung
- Conducted research on the efficacy of estimating safe stopping distances and times for ABB industrial robots with machine learning methods
- Programmed and conducted physical testing on industrial robot systems to assess their stopping behavior, working with industry advisors to align research outcomes with international safety standards

Mechanical Engineering Intern, GreenbergFarrow Architecture New York, NY May 2022 – Nov. 2023

- Performed design and analysis of real-world Mechanical, Electrical and Plumbing systems across a variety of building types
- Executed coordination efforts across engineering and architecture disciplines in Autodesk BIM360 both within the firm and with outside partners. Attended virtual coordination meetings with clients and project subcontractors.

Data Analytics Intern & Analyst, ParkLexington Advisors Conshohocken, PA July 2019 - Mar. 2020

- Collaborated with other Analysts and AbleNoser Consulting Group to build a product that provides insight into the Implicit and Explicit costs associated with trading securities
 - Developed data visualization and cleansing tools in Excel using Visual Basic for Applications to provide firm-specific demonstrations of trading costs to prospective clients
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Additional Skills and Qualifications

Languages: English (Native), German (B2 / Conversational)

Programming Languages: MATLAB & Simulink (Advanced), Python (Advanced - PyTorch, Matplotlib, Numpy), C (Intermediate), LaTeX (Intermediate), Visual Basic for Applications (Intermediate), RAPID (Intermediate), C++ (Beginner)

Technical Software: Autodesk Revit (Advanced), Microsoft Excel (Advanced), AutoCAD (Intermediate), Autodesk Inventor (Intermediate), ABB RobotStudio (Intermediate)

Music: 16 year of classical piano experience, classically trained tenor vocalist

Academic and Professional Honors

Deutschlandstipendium Scholarship Recipient (Sponsors Merck KGaA & TU Darmstadt), 2022-2023

Graduate Teaching Assistantship – Physics Department, Virginia Tech, 2021-2022

Fulbright ETA Fellowship in Germany, 2021 Alternate

Research Assistant, Hamilton College Surface Physics Laboratory, 2020-2021

Delta Phi Fraternity Nottingham Leadership Scholarship, Spring 2019 Recipient